Mace Analysis: Machine Learning Mingles with the Crowd

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Abstract

This study analyses the predictability of musical preferences of the audience at a MACE[1] concert, with the aim of developing a model capable of predicting which guest artist the audience most wishes to see perform. Through data collected via a survey on participants' habits and characteristics, we divided the audience into four macro-categories (Trap, Classic Rap, Indie Pop, and Alternative/Electronic) based on their musical preferences, and evaluated several machine learning algorithms, including Logistic Regression, Gradient Boosting, and Random Forest. To ensure model robustness and handle data imbalance, we employed advanced pre-processing techniques, oversampling via SMOTE, and hyperparameter optimisation through stratified cross-validation.
A key contribution of this work lies in the analysis of *feature importance*: by querying the models, we demonstrate that aggregating minority response categories (with importance below 1.5%) into an "Other" class reduces statistical noise and improves predictive performance. The results show that audience preferences do not depend on a single factor alone, but rather on the interaction of various individual characteristics and listening habits. Furthermore, the study highlights which pieces of information are most significant for preference prediction, suggesting how to optimise data collection in future surveys to improve both participant experience and predictive effectiveness.

1 Introduction

Understanding audience expectations and preferences is a crucial element in the live music industry. Anticipating spectators' desires not only improves the overall event experience, but also provides useful insights for understanding and analysing audience preferences. Consequently, it is fundamental to investigate the demographic, behavioural, and listening characteristics that drive such preferences.

***Research Question**: Is it possible to accurately predict the musical preferences of a heterogeneous audience based on socio-demographic and behavioural variables collected in real time? And to what extent can dataset refinement through feature engineering techniques improve the predictive accuracy of classification models?*

Our research team physically attended the highly anticipated MACE concert, held in Milan on 30 April 2026, experiencing first-hand an extraordinary event both musically and visually.



Figure 1: The research team at the MACE concert.



Figure 2: Spectators filling in the pre-show questionnaire.

To capture the real expectations of the spectators immersed in the pre-show atmosphere, we directly interacted with the crowd by distributing QR codes for an instant digital survey. This on-site data collection approach provided an authentic snapshot of the event population, gathering primary data on musical habits, personal characteristics, and, most importantly, the explicit desire to see certain guest artists perform.

Specifically, the questionnaire was structured into three sections:

[1] Mace (Simone Benussi) is a renowned Italian DJ and record producer, famous for his collaborations spanning the urban, rap, and indie-pop national scenes.

The first section investigated the socio-demographic and behavioural characteristics of the respondents: age, gender, geographical origin, prevailing clothing style, and frequency of concert attendance.
The second section focused on listening habits and the relationship with the artist: preferred music genre, number of years of regular listening to Mace, favourite album, and ways of attending the event.
Finally, the third section explored the experiential expectations and motivations of the participants: through a 1-5 Likert scale, three dimensions were investigated – the search for energy and moshing, the desire for a psychedelic journey, and the search for emotions and singing at the top of one's lungs – which revealed distinct experiential profiles. The final question, *"Which artist do you hope to see tonight?"*, constituted the target variable of the model, allowing us to map spectators' preferences onto the concert's guest artists.

This set of demographic, behavioural, experiential, and musical preference variables provided a rich and articulated data basis for developing predictive models, enabling the classification of the audience and the prediction of their general musical expectations.
The ultimate goal is to analyse to what extent such information allows predicting audience preferences and to understand which factors are most strongly associated with spectators' choices.

2 Exploratory Data Analysis

2.1 Demographic and Geographic Distribution

The sample consists of 319 respondents. The age analysis (Figure 3) reveals a structure with a clear positive skew. The audience is overwhelmingly composed of young adults, with an absolute peak around age 20 (74 observations). Interest in the artist consolidates in the post-adolescent and university range (18-24 years), then gradually fades after 26, leaving room for a physiological "long tail" of historical fans or cross-over listeners.

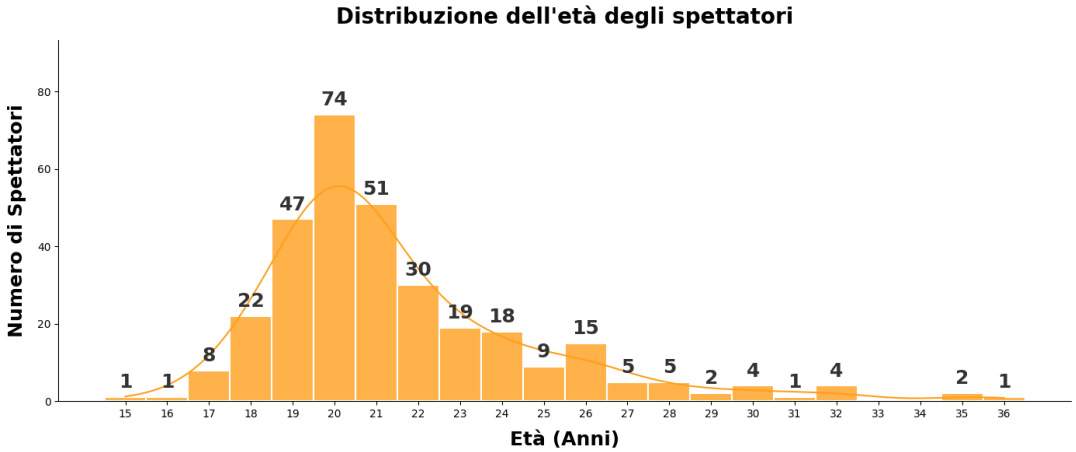


Figure 3: Density curve of the age distribution of spectators.

A detailed look at the absolute frequencies (Figure 3) confirms the impact of the 19-21 age bracket, which alone gathers 172 individuals (over 53% of the total sample).

From a territorial perspective (Figure 4), the sample shows a marked concentration in the regions of Northern and Central Italy. This distribution can be attributed both to the greater geographical accessibility of the event and to the larger presence, in these areas, of an audience interested in the musical genres represented by the artist.

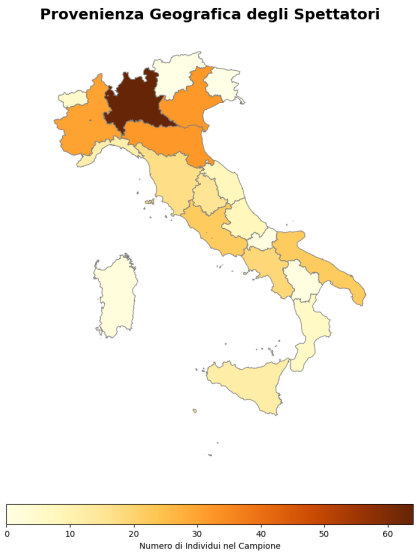


Figure 4: Geographical origin of spectators.

Table 1: Count by Region

Region	**N. Spectators**
Lombardy	64
Emilia Romagna	32
Veneto	32
Piedmont	30
Lazio	22
Apulia	22
Campania	19
Tuscany	17
Umbria	15
Sicily	12
Other	*54*
Total	**319**

The absolute data, summarised in Table 1, reveal a clear prevalence of Lombardy, which with 64 spectators records double the attendance of the second-placed regions (Emilia Romagna and Veneto, both at 32). A surprising and noteworthy figure is the strong response from Apulia (22 individuals) and Campania (19). Although far from the northern centre, they exceed or equal closer central regions such as Tuscany (17). Considering the greater distance from the event location, this suggests a significant ability to attract audiences from these two regions, although such distribution may be influenced by the presence of out-of-town students and workers residing near Milan.

2.2 Gender Composition and Musical Affinity

The sample shows a majority female participation (Table 2). Specifically, the raw data report 182 women and 133 men. The presence of 4 individuals identifying as "Other" confirms the inclusive nature of the audience. This demographic breakdown is particularly significant when compared with traditional stereotypes; the live event thus configures itself as a fluid and aggregating cultural space, capable of overcoming old demographic barriers and attracting a modern, heterogeneous, and deeply diversified fan base.

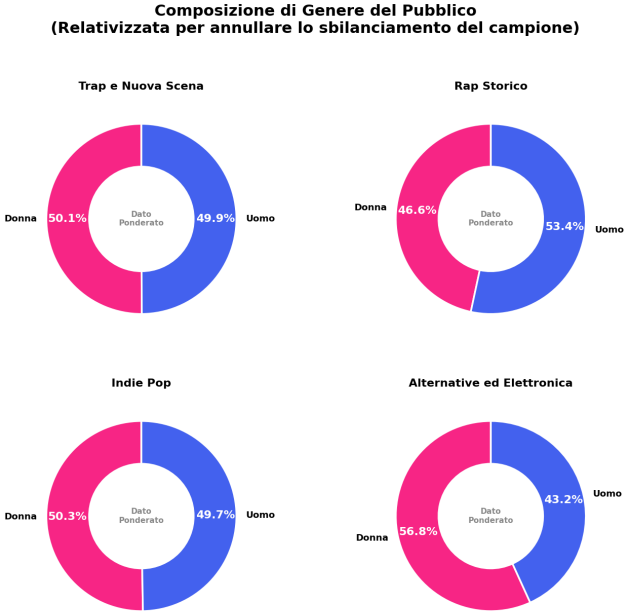


Figure 5: Donut chart of gender composition weighted by musical cluster

Table 2: Absolute frequencies by Gender

Gender	**N. Absolute**
Female	182
Male	133
Other	4
Total	**319**

In the analysis of musical preferences by gender, to reduce possible sampling biases, the cross-tabulation between respondents' gender and preferred macro-genre was analysed using relative weights. This weighting procedure allowed obtaining a more balanced representation of preferences within the different clusters (Figure 5). Although Trap and Classic Rap are stereotypically considered male-oriented genres, the massive female presence (57% of the total sample)

demonstrates a complete overcoming of gender barriers in the consumption of contemporary urban music. This aspect was confirmed by the weighting of the charts, which brought the Rap and Trap slices close to a perfect 50-50 balance.

2.3 Geographic Analysis by Musical Genre

The exploration of the geographic distribution disaggregated by preferred musical genre (visualised through the juxtaposition of choropleth maps and pie charts) represents a crucial node of the descriptive analysis. The aim of this breakdown in absolute values is twofold: on the one hand, it allows mapping the real territorial rooting of different musical scenes; on the other, by focusing on actual volumes rather than percentages alone, it prevents perceptual distortions caused by sample imbalance, revealing the true specific weight of each genre on the national territory.

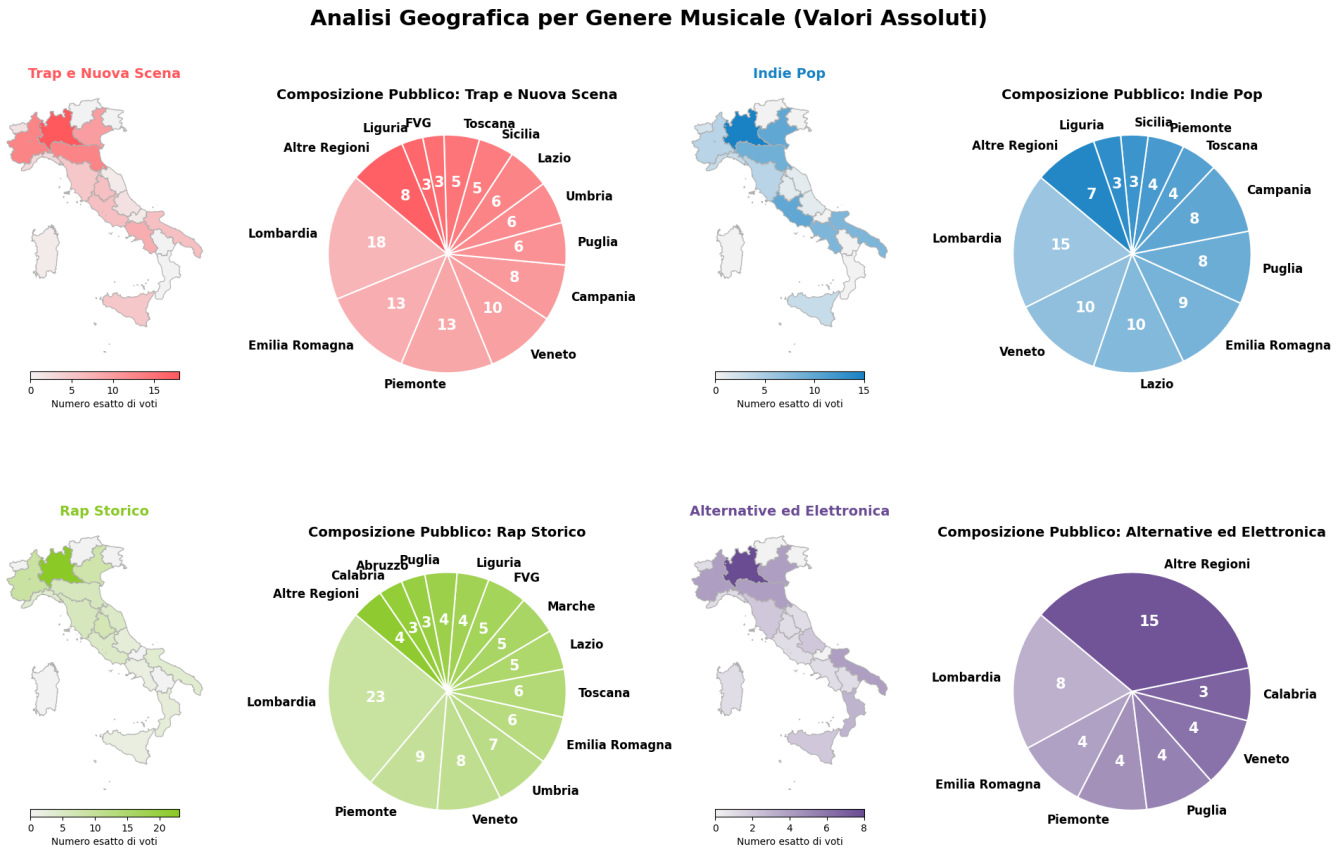


Figure 6: Geographic density and breakdown by musical macro-genre.

However, scaling absolute values to the size of the regional sample is essential for interpretation. This perspective reveals much more precise geographic patterns:

- **Indie Pop:** Shows an exceptional penetration in Central-Southern Italy. Despite the higher volumes in the North, in Lazio (10 out of 22) and Campania (8 out of 19) this genre approaches or exceeds 40% of local preferences.
- **Classic Rap:** Strongly rooted in the North-West (in Lombardy and Piedmont it represents the relative majority), its incidence drops sharply moving down the peninsula.
- **Trap and New Scene:** Confirms itself as the most widespread cluster. Although recording its absolute peak in Lombardy (18 out of 64, about 28%), its true proportional strength emerges in Emilia Romagna and Piedmont, where it captures over 40% (respectively 13 out of 32 and 13 out of 30) of the local basin.
- **Alternative and Electronic:** Reaffirms its nature as a transversal niche, with a marginal but constant incidence (maximum 1-4 individuals) distributed homogeneously across the national territory.

In summary, the relative analysis scales down the absolute weight of Northern Italy and reveals a very high density of *Indie Pop* and *Trap* listeners in the Centre-South. Besides demonstrating a strong propensity for mobility of these fans to attend the event, the data suggest the presence of a southern market with still largely untapped potential for the live industry.

2.4 Behavioural Profiling and Expectations

The motivations that drive the audience to attend the event vary according to musical background. By visualising the three main expectations on a 1-5 scale on a polar graph (Figure 7), the "psychological profile" of each musical cluster emerges, compared with the dashed line representing the global average.

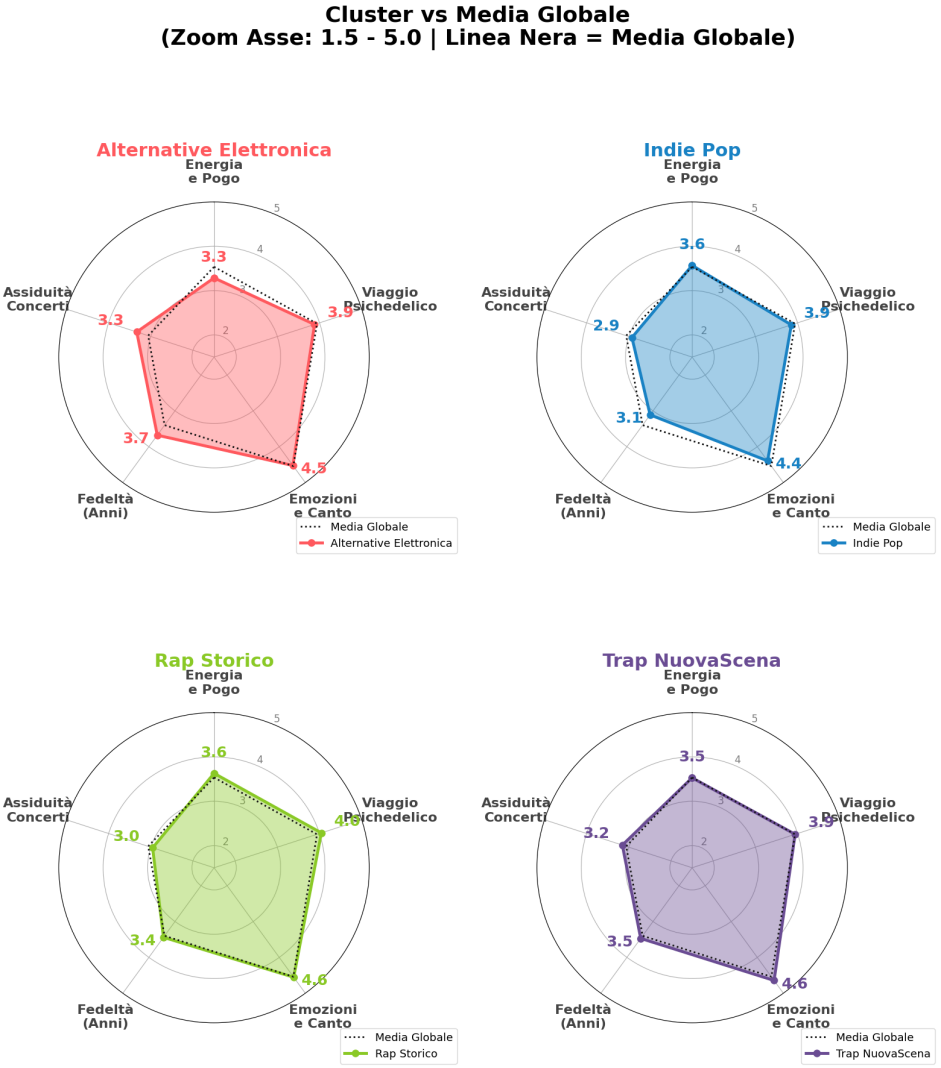


Figure 7: Radar chart of emotional and behavioural expectations of fans.

Distinctive behavioural patterns emerge that explain the polarisation of preferences:

- **Alternative/Electronic:** Records the highest values overall for *Years of Loyalty* (3.7) and *Concert Attendance* (3.3), denoting a mature and consistent fan base over time. On the other hand, the live approach is much more subdued: reaching the minimum values in both *Emotions and Singing* (4.3) and *Energy and Mosh* (3.3). *Psychedelic Journey* (3.9) aligns perfectly with the global average.
- **Indie Pop:** Records the minimum values in *Concert Attendance* (2.9) and *Years of Loyalty* (3.1), outlining fans with occasional live attendance and less historicised attachment. It stands out for the physical component *Energy and Mosh* (3.6 vs 3.5 global average), while interest in the *Psychedelic Journey* remains aligned with the rest of the respondents.
- **Classic Rap:** Presents an audience with very low *Concert Attendance* (3.0) and *Years of Loyalty* below average (3.4 vs 3.5 general). It absolutely dominates the *Emotions and Singing* category, pushing the vertex to the maximum peak of the entire chart (4.6). Moreover, they are the only cluster to exceed the average in *Psychedelic Journey* (4.0 vs 3.9 global). With *Energy and Mosh* (3.6), slightly above average, the profile of a fan who attends few concerts but with a deeply emotional experience emerges.
- **Trap and New Scene:** Records perfectly median values for *Energy and Mosh* (3.5), *Psychedelic Journey* (3.9) and *Emotions and Singing* (4.6). Even *Years of Loyalty* (3.5) and *Concert Attendance* (3.2) sit exactly on the standard of the entire sample. This group represents the true balancing point of the whole survey: the

absence of marked polarisations indicates that the approach of these fans reflects the habits and expectations of the standard listener.

In summary, the Radar Chart demonstrates that artistic preference is deeply rooted in the experiential approach to live music. However, the partial overlap of some profiles (such as Trap mirroring the global average) and the nuanced nature of these responses explain the complexities encountered by classification algorithms, justifying the moderate predictive performance due to strong individual variability.

3 Pre-processing and Feature Engineering

The preparation of the dataset required a careful phase of cleaning and logical transformation, essential to ensure stability and consistency for the subsequent Machine Learning models. The approach was articulated in the following fundamental steps:

3.1 Target Management and Clustering of Musical Genres

The target variable had over 20 labels for 319 observations, causing two concatenated problems: *class sparsity* (artists with few votes) and *structural overfitting* (memorisation of patterns in the sample).
To overcome this problem, we adopted a technique of **semantic clustering guided by domain knowledge**, preferred to an automatic approach (e.g. *K-means*) due to the absence of vector features of the artists that would allow training such a model.
Through analysis of stylistic affinities and prevailing *sentiment*, the artists were thus grouped into four macro-genres:

- Classic Rap
- Trap/New Scene
- Indie/Pop
- Alternative/Electronic.

This reduction brought the problem back to a statistically tractable multi-class classification (4 classes), while preserving the semantic nature of the expressed preferences.

3.2 Handling Missing Values and Dimensionality Reduction

During the initial cleaning, missing observations (NaN) were treated through standard methodologies: substitution with the string "Unknown" for qualitative variables and interpolation with the median value for quantitative ones.
In parallel, we proceeded to a first aggregation phase of categories for some dispersed *features*:

- Regions of origin were reduced to macro-areas (North, Centre, South and Islands).
- Open-ended answers regarding descriptions of the artist (Mace) were restricted to the four most frequent adjectives ("Psychedelic", "Emotional", "Energetic", "Technical"), enclosing the other textual variations in the generic category "Other".

3.3 Encoding of Ordinal Variables

When transforming textual data into numerical format, special attention was paid to the logical nature of each variable to avoid confusing the algorithms. *Features* endowed with an intrinsic hierarchy or progression were treated as **ordinal variables**.
In particular, the continuous age variable was discretised into three textual bands ("Under 20", "20-25" and "Over 25") and subsequently mapped with progressive integer values (0, 1, 2); similarly, the years of regular listening to the artist were mapped on an ordinal scale (from 0 to 3).

This approach ensured the preservation of relational information (e.g. a value of 2 is logically and mathematically greater than 0), helping tree-based models to make optimal splits.

3.4 Resolving False Ordinality (One-Hot Encoding)

For all remaining nominal variables (such as region, clothing style, beverage consumed), we intentionally avoided the use of standard *Label Encoding* to prevent the problem of **false ordinality**. Being strictly **nominal** variables, assigning sequential integer values would have induced mathematical models to deduce non-existent hierarchies or compute fictitious "distances" between categories.
To these variables we applied a rigorous **One-Hot Encoding** approach, creating independent binary variables (*Dummy Variables*) for each response modality. Although expanding the dimensionality of the predictive space, this transformation guaranteed the model a correct data architecture on which to train without any logical *bias*.

4 Model Selection and Training

Once the *Feature Engineering* was completed, the resulting dataset was prepared for the machine learning phase. The goal was to train different *Machine Learning* algorithms while ensuring maximum methodological correctness, to avoid over-optimistic performance estimates or biases due to design errors.
To evaluate the models' generalisation ability on unseen data, the dataset was initially split into two subsets: a *Training Set*, containing 80% of the observations, used exclusively for learning, and a *Test Set*, representing the remaining 20% of the observations, isolated and kept solely for performance evaluation.

4.1 Class Balancing via SMOTE

The exploratory analysis highlighted a moderate imbalance in the frequencies of the Target variable. The majority class (Trap/New Scene) had more than double the volume of observations compared to the minority class (Alternative/Electronic, consisting of only 36 individuals). In such scenarios, machine learning algorithms tend to develop a strong bias toward the dominant classes, because the standard cost function is optimised to maximise overall accuracy, neglecting the less represented classes.
To address the problem, the analysis team posed a fundamental methodological question: *is it statistically safer to generate synthetic data or to mathematically penalise errors?* Two parallel experiments were therefore conducted to compare two different balancing philosophies:

1. **Synthetic Sampling (SMOTE)**: The algorithm creates new artificial examples for the less represented classes by combining the characteristics of real instances close to each other. In this way, it enriches the dataset without merely duplicating existing examples, helping the model better recognise minority classes. The theoretical risk in this case was *overfitting*: generating dozens of synthetic profiles from a very small real set could have amplified statistical noise.

2. **Algorithmic Balancing (Class Weights)**: No data alteration. The algorithm's cost function is modified (e.g. `class_weight='balanced'`) by penalising more severely errors made on the minority class.

To settle the question rigorously, the two approaches were evaluated on the three candidate models (Logistic Regression, Random Forest and Gradient Boosting) using a single `train-test` split (80

Table 3: Comparison of SMOTE vs Class Weight on base models

Model	SMOTE (%)	Class Weight (%)
Logistic Regression	40.62	32.81
Random Forest	43.75	35.94
Gradient Boosting	37.50	34.38

Contrary to initial concerns, **SMOTE** proved to be the clearly superior approach on all three models, with a particularly marked advantage on Random Forest (43.75% vs 35.94%). From a statistical perspective, this result indicates that individuals in the minority class had sufficient behavioural consistency to make spatial interpolation effective. SMOTE did not amplify statistical noise, but strengthened a real behavioural signal, helping the algorithm draw clearer decision boundaries. SMOTE was therefore confirmed as the standard balancing technique for the entire project.

Crucial for the scientific validity of the experiment was the application of SMOTE **exclusively on the Training Set**. Balancing the entire dataset before splitting would have introduced a serious methodological error known as *Data Leakage*, "contaminating" the *Test Set* with synthetic information derived from training data and irreparably invalidating the final metrics. For this reason, in all subsequent phases of the analysis, SMOTE was applied strictly after data splitting and, in the case of Cross-Validation, within each training fold using dedicated pipelines.

4.2 Candidate Algorithms

To identify the most suitable architecture for deciphering the often non-linear and multifaceted relationships between respondents' lifestyles and their musical preferences, three algorithms based on different theoretical paradigms were selected:

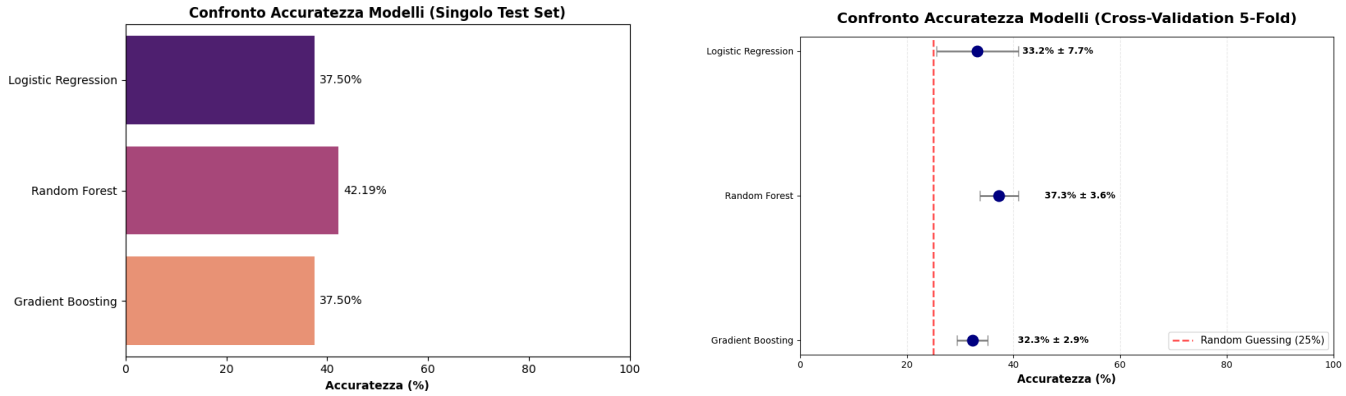
- **Logistic Regression**: implemented as a linear *baseline* model. Although offering excellent interpretability, its linear nature makes it less suitable for capturing complex interactions (e.g. specific combinations of age, origin and clothing, etc.).

- **Gradient Boosting**: a sequential *Ensemble* method. It builds decision trees in series, where each new tree attempts to minimise the residual error of the previous one. It is an extremely powerful algorithm, but particularly sensitive to the risk of *overfitting* on small-sized datasets.

- **Random Forest**: a parallel *Ensemble* method based on the *Bagging* technique. It generates multiple independent decision trees on random subsamples of data and *variables*, aggregating the final predictions through a majority voting mechanism. This structure makes it intrinsically more robust and resistant to statistical noise.

4.3 Baseline Results and Overfitting Detection

The three models were initially trained on the encoded variables to establish a performance baseline (*baseline evaluation*), a preparatory step for the subsequent rigorous validation and hyperparameter optimisation phase. To ensure proper convergence of Logistic Regression, the data were standardised using `StandardScaler`; this transformation was applied to all models for workflow uniformity, although tree-based algorithms are scale-invariant.



(a) Single split

(b) Cross-Validation 5-fold

Figure 8: Comparison of model accuracies: (a) evaluation on a single Test Set, (b) evaluation via 5-fold Cross-Validation with standard deviations.

The results of the initial accuracy comparison, reported in Figure 8a, showed a predominance of Random Forest, which achieved an accuracy of **42.19%** on the Test Set, surpassing Logistic Regression (37.50%) and Gradient Boosting (37.50%).

However, evaluating a model's *performance* on a single *Test Set* (in our case, 20% of the sample, about 80 observations) can be misleading. On small-sized samples, statistical variance is high: an algorithm might achieve excellent scores purely by chance (the so-called *lucky split*), performing well on that specific set but failing to generalise to future data. To ensure the solidity of the results, a more severe validation approach was adopted: Global Cross-Validation (Stratified K-Fold).

To compute the true average accuracy of the algorithms, the entire dataset was subjected to a **Stratified 5-fold Cross-Validation**. The sample was divided into 5 balanced subsets with respect to the Target classes; iteratively, 4 subsets were used for *training* and 1 for *testing*.
To prevent *Data Leakage* during the 5 validation cycles, a *Pipeline* was built that recomputed SMOTE each time exclusively on the current 4 *training folds*. The choice to use 5 folds represents the ideal standard in this context, as it allows reducing the computational time of the Pipeline (e.g. with 10 folds it would double).
The results of the *Cross-Validation* highlighted fundamental discrepancies compared to the preliminary single-split tests.

Figure 8b compares the performance of the three models evaluated via Stratified 5-fold Cross-Validation. Each point represents the average accuracy obtained by each model, while error bars indicate the standard deviation of accuracies across the 5 folds, providing a measure of model stability. **Logistic Regression** has an average accuracy of 33.2%, but shows the highest variability (standard deviation of ±7.7%), with an accuracy range between 26.5% and 44.4%. This high variability suggests lower robustness of the linear model compared to tree-based approaches.

As already mentioned, **Random Forest** emerges as the model with the best overall performance, reaching an average accuracy of **37.3%** with a standard deviation of ±3.6%, hence with much more contained oscillations across the various *folds*).

Gradient Boosting, compared to the performance obtained with the single split, recorded a reduction in the overall average accuracy, in this case equal to **32.29%**, the lowest among the models considered, demonstrating a pronounced tendency toward *overfitting*. However, it should be noted that it stands out for its greater stability (with a standard deviation of ±2.9%).

From this initial analysis it therefore follows that Random Forest, although not the most stable algorithm in absolute terms, offers the best compromise between accuracy and robustness, able to decipher general behavioural rules without memorising exceptions.

However, to have a benchmark during optimisation and not to neglect the potential of Gradient Boosting (known in the literature for its ability to achieve high performance with appropriate tuning), we chose to explore both ensemble algorithms in depth. Below we report the logical flow we followed in training the two models and the results that emerged.

5 Optimisation and Comparison of Algorithms

After identifying the most promising models, we conducted an in-depth analysis on both ensemble algorithms (Random Forest and Gradient Boosting) to evaluate their potential and select the most suitable for the problem at hand.

5.1 Gradient Boosting

Gradient Boosting was initially evaluated on the single Test Set with default parameters, achieving, as already mentioned, an accuracy of **37.50%**. Subsequently, we proceeded with hyperparameter optimisation of the model via `GridSearchCV` with 5-fold cross-validation. To ensure correct handling of imbalance and data standardisation without incurring data leakage, the entire procedure was encapsulated in a pipeline that integrates SMOTE and StandardScaler. The search grid explored combinations of three key parameters: `n_estimators` (50, 100, 150), `max_depth` (2, 3, 5) and `learning_rate` (0.01, 0.05, 0.1). The optimal parameters identified are reported in Table 4.

Table 4: Optimal parameters for Gradient Boosting (cv=5)

Parameter	Value	Meaning in the model
`n_estimators`	50	Prudent value for small datasets; additional trees (> 50) did not improve CV mean
`max_depth`	2	Extremely shallow trees to prevent the model from learning training-set-specific patterns
`learning_rate`	0.01	Very low learning rate to reduce the impact of each tree and favour regularisation

The optimised model achieved an accuracy of **42.19%** on the Test Set, recording an improvement of almost 5 percentage points over the base model. To verify the generalisability of the result, a 5-fold Cross-Validation was conducted, which returned an average accuracy of **38.26%**. The gap of about 4 percentage points between the Test Set accuracy (42.19%) and the Cross-Validation average (38.26%) suggests a residual tendency to overfitting, since the model likely learned patterns present in that 20% of data used in testing.

With the model now calibrated on optimal parameters, we questioned the real contribution of marginal variables to predictive power: how many of them constituted signal and how many, instead, simple statistical noise? To answer, we extracted the feature importance ranking (Figure 9), which returned a strongly unbalanced picture.
To verify the intuition derived from the feature importance analysis, according to which most of the predictive power was concentrated on few variables, we conducted a pooling experiment: response modalities with a weight below 1.5% were collapsed into an *Other* category, reducing the feature space from 36 to 20 columns.

After pooling the variables, we deemed it appropriate to perform a new search for optimal parameters via grid search (in Table 5) based on the new dataset, to guarantee the best possible results.

Table 5: Optimal parameters for Compacted Gradient Boosting (cv=5)

Parameter	Value	Meaning in the model
`n_estimators`	50	Prudent value for small datasets; additional trees (> 50) did not improve CV mean
`max_depth`	3	Increase in maximum depth compared to the pre-pooling model
`learning_rate`	0.05	Higher learning rate than before

Retraining the model with the new parameters on the new dataset, we observed a further improvement in performance: the cross-validated accuracy reached **45.10%**. It emerges that simplifying the model not only improves its

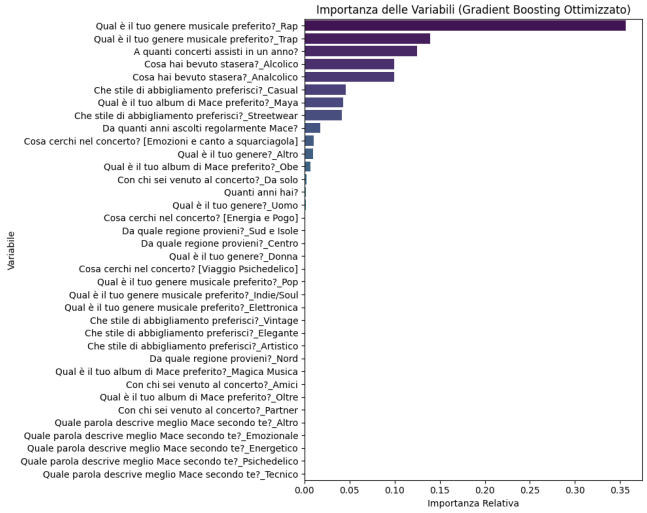


Figure 9: Feature Importance – Optimised Gradient Boosting

The genres *Rap* (35.7%) and *Trap* (13.9%) confirm themselves as the most influential predictors, followed by the number of annual concerts (12.5%) and beverage choice (about 10%). While highlighting a clear predominance of variables related to musical genre — which alone exceed 49% of the overall predictive weight — the model also draws on other behavioural factors, such as consumption habits and frequency of attendance at live events. This suggests that audience preferences are not reducible to the stylistic label alone, but are the result of a more complex interplay of elements related to lifestyle and individual experiences.

interpretability, but also makes it more precise in predictions.

The performance analysis of the compacted Gradient Boosting, conducted through the classification report (Table 6) and the confusion matrix (Figure 10), offers a detailed picture of the model's capabilities.

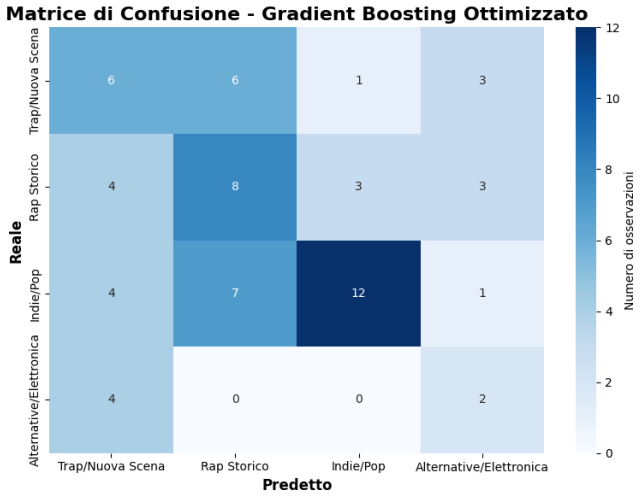


Figure 10: Confusion matrix – Compacted Gradient Boosting

Table 6: Evaluation metrics – Compacted Gradient Boosting

Class	**Prec.**	**Rec.**	**F1**	**Supp.**
Trap/New Scene	0.33	0.38	0.35	16
Classic Rap	0.38	0.44	0.41	18
Indie/Pop	0.75	0.50	0.60	24
Alternative/Electronic	0.22	0.33	0.27	6
Weighted Avg	0.49	0.44	0.45	64

The *Indie/Pop* class stands out as the best predicted, with 12 out of 24 observations correctly classified (50%) and an F1-score of 0.60, driven by excellent precision (0.75) (a sign that when the model predicts this class, it does so with high confidence and accuracy).
The *Trap/New Scene* and *Classic Rap* classes show intermediate profiles (F1-scores of 0.35 and 0.41, respectively), with correct classification rates of 38% (6/16) and 44% (8/18).
For the *Alternative/Electronic* class, the model achieves a recall of 33% (2/6) and a precision of 0.22, with an F1-score of 0.27.

The confusion matrix reveals that the main confusions involve *Trap/New Scene*, which is mostly confused with *Alternative/Electronic* (6 errors), indicating a persistent overlap between these two clusters. *Indie/Pop* and *Classic Rap* show a pattern of mutual confusion (5 errors in both directions), suggesting similarity in the behavioural profiles of fans of these two genres.

The weighted average (precision = 0.49, recall = 0.44, f1-score = 0.45) confirms an overall sufficiently satisfactory predictive power, but to further improve performance, collecting additional data for the *Alternative/Electronic* class remains the priority intervention.

The experiment demonstrates that Gradient Boosting, if properly calibrated and regularised, can achieve competitive performance; however, its sensitivity to sample size and tendency to concentrate predictive weight on few variables still limit its reliability in this context.

5.2 Random Forest

To ensure a fair comparison with Gradient Boosting and maintain methodological consistency, Random Forest was subjected to the same hyperparameter optimisation procedure. The base model, which on the single Test Set had achieved an accuracy of 42.19% and in Cross-Validation an average of 37.30%, was then optimised via `GridSearchCV` with 5-fold cross-validation, using a pipeline with SMOTE to avoid data leakage. The search grid was designed according to literature recommendations for small datasets: `n_estimators` (100, 200, 300) to balance predictive power and computational cost; `max_depth` (5, 10, None) to explore a wide spectrum of tree depths; `min_samples_leaf` (1, 2, 4) to regularise model complexity; and `criterion` (gini, entropy) to evaluate both standard impurity metrics. Note that the Random Forest search grid is deliberately more extensive than that defined for Gradient Boosting; these choices reflect the different natures of the two algorithms: Random Forest benefits from many deep trees, while Gradient Boosting favours shallow trees with low learning rates.
The parameters that maximised the average Cross-Validation accuracy are reported in Table 7.

Table 7: Optimal parameters for Random Forest

Parameter	Value	Meaning in the model
`criterion`	entropy	Impurity measure based on entropy, preferred to gini in this context
`max_depth`	10	Sufficient depth to capture complex interactions without overdoing
`min_samples_leaf`	4	Moderate regularisation: avoids too small leaves that would lead to overfitting
`n_estimators`	200	Number of trees that balances stability and computational cost for the dataset

In this case, the optimised model achieved an average Cross-Validation accuracy of **40.13%**. On the single Test Set, the accuracy was **39.06%**, in perfect alignment with the cross-validated estimate (difference of about 1 percentage point). Unlike Gradient Boosting (which had a gap of about 4 percentage points between Test Set and Cross-Validation, a sign of an overfitting tendency), Random Forest shows substantial consistency between the two estimates, confirming its greater robustness and generalisation ability.

Similarly to what was done for Gradient Boosting, we proceeded to extract feature importance also for Random Forest (Figure 11), to understand which variables drove the model's predictions.

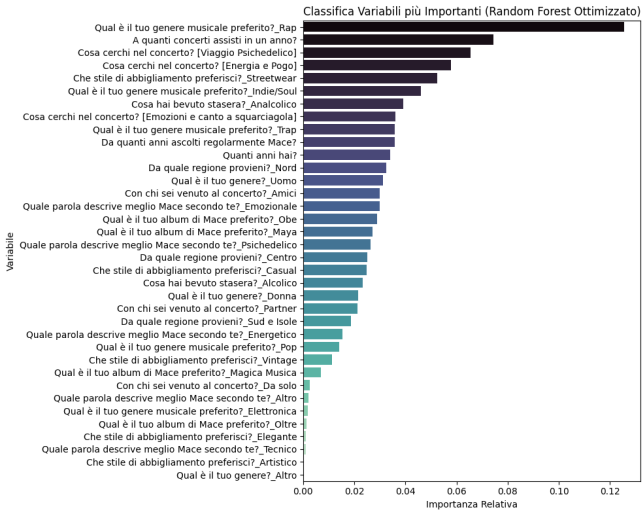


Figure 11: Feature Importance – Optimised Random Forest

The resulting distribution appears notably more balanced than that observed for Gradient Boosting: the musical genre *Rap* (12.6%) and the number of annual concerts (7.4%) stand as the main variables, followed by experiential expectations (*Psychedelic Journey*, 6.5%; *Energy and Mosh*, 5.8%) and *Streetwear* clothing style (5.2%). This distribution reflects Random Forest's ability to draw on a broader range of behavioural signals, without excessively concentrating on a few dominant variables, unlike Gradient Boosting, where Rap and Trap alone exceeded 49% of the predictive weight.

Response modalities with a weight below 1.5% were grouped into a macro-category *Other*, reducing the dimensional space from 36 to 31 columns (removing 5 variables, corresponding to minority response categories such as clothing

styles, niche musical preferences, and event participation modes). The model retrained on the compacted dataset maintained the accuracy unchanged at **39.06%**.

Here too, we reran the optimal parameter search on the new pooled dataset (in Table 8).

Table 8: Optimal parameters for compacted Random Forest

Parameter	Value	Meaning in the model
`criterion`	entropy	Impurity measure based on entropy, preferred to gini here as well
`max_depth`	10	Sufficient depth to capture complex interactions without overdoing, unchanged from previous case
`min_samples_leaf`	1	Even stricter regularisation
`n_estimators`	200	unchanged

The performance analysis of the compacted Random Forest, conducted through the classification report (Table 9) and the confusion matrix (Figure 12), offers a detailed picture of the model's capabilities and limitations, allowing a direct comparison with what emerged for Gradient Boosting.

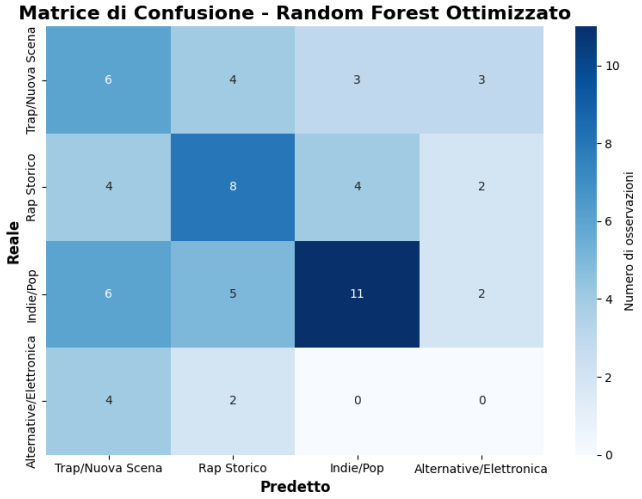


Figure 12: Confusion matrix – Compacted Random Forest

Table 9: Evaluation metrics – Compacted Random Forest

Class	Prec.	Rec.	F1	Supp.
Trap/New Scene	0.30	0.38	0.33	16
Classic Rap	0.42	0.44	0.43	18
Indie/Pop	0.61	0.46	0.52	24
Alternative/Electronic	0.00	0.00	0.00	6
Weighted Avg	0.42	0.39	0.40	64

The model achieves an average accuracy of **44.71%**, also improved compared to the pre-pooling case.

The *Indie/Pop* class emerges as the best predicted, with 11 out of 24 observations correctly classified (46%) and an F1-score of 0.52, driven by good precision (0.61), a sign that the model effectively recognises the profiles of this cluster, with a limited false positive rate.
The *Classic Rap* class shows a solid and balanced profile (F1-score = 0.43), with 8 out of 18 correct observations (44%) and precision and recall of 0.42 and 0.44, respectively, indicating a symmetric behaviour of the model on this class.
The *Trap/New Scene* class, on the other hand, presents an F1-score of 0.33, with 6 out of 16 correct observations (38%) and a recall (0.38) higher than precision (0.30), suggesting a tendency of the model to overestimate this class, misclassifying as Trap examples belonging to other categories.
The *Alternative/Electronic* class represents the most critical point: despite the 6 observations present in the Test Set, the model does not correctly classify any of them, with metrics equal to zero on precision, recall, and F1-score. This behaviour is likely attributable to the small sample size (only 6 observations), which did not provide the model with enough examples to learn a distinctive behavioural profile, and to the possible overlap with other clusters.

The main confusions are concentrated along three directions: *Trap/New Scene* is mainly confused with *Alternative/Electronic* (4 errors) and with *Indie/Pop* (3 errors), suggesting an overlap between the profiles of fans of these genres. *Indie/Pop* and *Classic Rap* show a pattern of mutual confusion (5 errors in both directions), indicating significant similarity in the behavioural profiles of fans of these two genres. Finally, *Alternative/Electronic* is distributed among the other classes, with 4 errors toward *Trap/New Scene* and 2 toward *Classic Rap*.

The weighted average (precision = 0.42, recall = 0.39, f1-score = 0.40) confirms a solid overall predictive power, in line with cross-validated estimates and with the results of Gradient Boosting. However, the model's inability to recognise even a single example of the *Alternative/Electronic* class represents a significant limitation, suggesting the need to collect additional data for this class in order to improve the model's ability to discriminate this cluster.

5.3 Final Comparison

The results presented so far outline a picture of the substantial differences between the two models, emerging on several fronts.

Regarding the pre-pooling phase, after optimisation Random Forest distinguished itself primarily by greater stability: while Gradient Boosting showed a gap of about 4 percentage points between Test Set accuracy (42.19%) and cross-validated mean (38.26%) — a sign of a residual overfitting tendency — Random Forest showed substantial consistency between the two estimates (39.06% vs 40.13%), confirming its robustness and generalisation ability.

This different nature is also reflected in the distribution of predictive weight: Gradient Boosting concentrates over 49% of the importance on the sole variables *Rap* and *Trap*, while Random Forest distributes the weight more uniformly across a broader range of features (musical genre, annual concerts, experiential expectations, and clothing style), suggesting a greater ability to draw on diversified behavioural signals.

More interesting and controversial are the results related to variable pooling: although we observed an improvement in the performance of both algorithms (accuracy for both around 45%), the overall highest score, both in terms of accuracy and the other metrics analysed, was achieved by Gradient Boosting (although in percentage terms Random Forest improved more).

The overall comparison between the two models is summarised in Table 10. Although reaching a higher peak on the single Test Set (42.19% vs 39.06%), Gradient Boosting showed marked instability in Cross-Validation and lower predictive reliability, being more prone to overfitting. Random Forest, on the other hand, demonstrated greater robustness and better calibration of predictive probabilities, confirming itself as the most reliable model for the problem at hand.

Table 10: Final comparison between Random Forest and Gradient Boosting

Metric	Random Forest	Gradient Boosting
Average Accuracy (CV)	**40.13%**	38.26%
Accuracy (Test Set)	39.06%	**42.19%**
Accuracy after Pooling	44.71%	**45.10%**
Column Reduction	36 → 31	36 → 20

Given the substantial balance found in the performance of the two models, we decided to carry out an additional comparative analysis: the distribution of predictive confidence of the two models was analysed (Figures 13 and 14), which provides a measure of the robustness of predictions: the horizontal axis reports the level of certainty assigned to each prediction (0.25 = random, 1.0 = 100% certain), while the vertical axis indicates the frequency with which such levels occur in the Test Set.

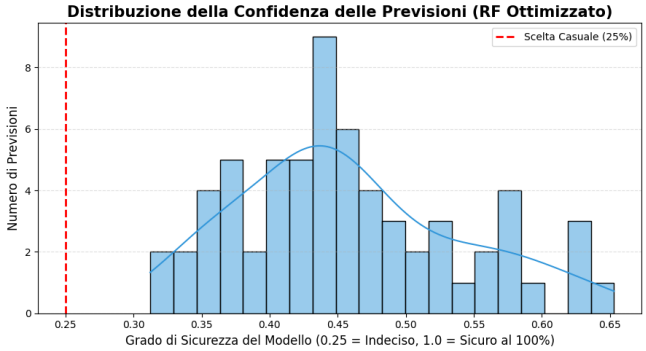


Figure 13: Distribution of confidence – Compacted Random Forest

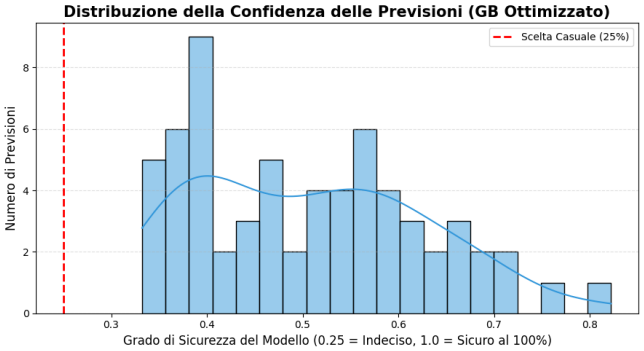


Figure 14: Distribution of confidence – Compacted Gradient Boosting

Random Forest (Figure 14) shows a confidence distribution that extends healthily between 0.30 and 0.65, approximating a bell-shaped curve with a density peak around 0.45. This behaviour is indicative of a robust model for a

complex multi-class problem: predictions reach safety peaks above 60% (a very solid margin compared to the 25% of *random guessing*), ensuring good discrimination between clusters while maintaining a realistic margin of systemic uncertainty.

Gradient Boosting (Figure 15), on the other hand, shows a slightly different dynamic: the distribution of prediction confidence is concentrated in a wider range (between 0.32 and 0.82), highlighting numerous predictions with confidence above 0.60. This indicates that the model alternates prudent decisions with much more convinced ones compared to Random Forest.

In conclusion, the challenge between the two algorithms does not decree an absolute winner, but confronts us with a clear analytical trade-off. On the one hand, Gradient Boosting proves to be the more "aggressive" choice: it reaches the maximum accuracy peak (45.1%) by creating an extremely compact model (only 20 variables), at the cost of strong decision polarisation and greater statistical instability. On the other hand, Random Forest confirms itself as the safer and more structured option: although giving up a few fractions of a percentage point in performance (44.7%), it guarantees marked stability between Cross-Validation and real Test, offering a more balanced and complete reading of the audience's facets.

Beyond the chosen model, the real triumph belongs to data quality: the experiment unequivocally demonstrates that reducing noise and strategically compacting variables is the fundamental key to maximising the predictive performance of any algorithm, thus suggesting that more streamlined and targeted questions should be asked in future questionnaires.